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ORIGINAL PAPER

Evaluation of a novel artificial intelligence-based screening system for diabetic retinopathy in community of China: a real-world study

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Abstract
Purpose To evaluate the performance of an AI-based diabetic retinopathy (DR) grading model in real-world community clinical setting.
Methods Participants with diabetes on record in the chosen community were recruited by health care staffs in a primary clinic of Zhengzhou city, China. Retinal images were prospectively collected during December 2018 and April 2019 based on intent-to-screen principle. A pre-validated AI system based on deep learning algorithm was deployed to screen DR graded according to the International Clinical Diabetic Retinopathy scale. Kappa value of DR severity, the sensitivity, specificity of detecting referable DR (RDR) and any DR were generated based on the standard of the majority manual grading decision of a retina specialist panel.
Results Of the 193 eligible participants, 173 (89.6%) were readable with at least one eye image. Mean [SD] age was 69.3 (9.0) years old. Total of 321 eyes (83.2%) were graded both by AI and the specialist panel. The κ value in eye image grading was 0.715. The sensitivity, specificity and area under curve for detection of RDR were 84.6% (95% CI: 54.6– 98.1%), 98.0% (95% CI: 94.3–99.6%) and 0.913 (95% CI: 0.797–1.000), respectively. For detection of any DR, the upper indicators were 90.0% (95% CI: 68.3–98.8), 96.6% (95% CI: 92.1–98.9) and 0.933 (95% CI: 0.933–1.000), respectively.
Conclusion The AI system showed relatively good consistency with ophthalmologist diagnosis in DR grading, high specificity and acceptable sensitivity for identifying RDR and any DR.
Translational relevance It is feasible to apply AI-based DR screening in community.
Precis Deployed in community real-world clinic setting, AI-based DR screening system showed high specificity and acceptable sensitivity in identifying RDR and any DR. Good DR diagnostic consistency was found between AI and manual grading. These prospective evidences were essential for regulatory approval.

Keywords Artificial intelligence · Diabetic retinopathy · Deep neural network · Fundus photography · Rreal-world study

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Introductions

Diabetic retinopathy (DR) is the most common sight-threatening complication in diabetes mellitus (DM) patients [1, 2]. China has the world's largest population of adults with DM which will continue to increase [3]. According to the national-wide surveillance, the estimated overall prevalence of diabetes reached to 10.9% among adults in China [4]. Such a big DM population is a potential vital pool of DR. Adequate quality of primary health care and systematic early screening in community has been the most important measures for preventing DR prior to vision loss [1, 5]. Although worldwide recommended at least annual screening in community diabetes care [6, 7], less than 50% of DM patients undergoes DR screening even in developed countries [8, 9]. Fewer than half of DR patients don't even know their status [10], especially in health resource limited remote areas.

Fundus image provides a feasible screening method for the detection of DR because of its cost-effective and noninvasive characters [11]. Artificial intelligence (AI)-based DR automated screening or grading program has innate advantage in automatically analyzing fundus photographs. Studies have reported AI program showed better diagnosis performance than ophthalmologist with more than 90% sensitivity and specificity in detecting referable DR (RDR) [12–14]. Such AI program is friendly to community where ophthalmic care resource is limited. DR screening can be integrated in primary health care by general practitioners or endocrinologists. AI-based DR screening inevitably offers potential to reduce the ophthalmologists' workload [15], increase the efficiency, sustainability and accessibility of DR screening programs worldwide.

Despite great advances had achieved, high accuracy of current AI algorithm was tested typically on retrospective laboratory data of selected high-quality images in silico. Before an algorithm can be implemented in the primary care process, its accuracy and safety have to be assured in real-world situation. As we know, validation of diagnostic algorithms in real-world settings such as primary health care site in community remains an emerging challenge and will be essential for future regulatory approval.

In light of these issues, we experimentally deployed one novel AI-based grading system for DR in a community clinic to assess its practical performance and prospectively determine the accuracy in DR screening among aged Chinese population in real-world setting.

Materials and methods

Study design

A prospective, cross-sectional real-world diagnosis validation trial of an AI-based DR screening system was designed in this study. Diabetes patients were prospectively recruited to clinic by community health care staffs every Monday, Wednesday and Friday to have free physical examination in a local primary clinic. The study was approved by the institutional review board of Henan Provincial Eye Hospital.

Participants

During December 2018 and April 2019, community citizens with diabetes managed in a local primary health care hospital, the second outpatient clinic of the affiliated institutions of Henan province, were mobilized in advance to attend the free AI-assisted DR screening program. Three primary health care staffs were responsible for checking the digital records of respondents to ensure their eligibility. Inclusion criteria were (1) diabetes information were recorded in the chronic disease management database; (2) $\geq$ 40 years old; (3) no previous diagnosis records of DR; (4) signed the informed consent. Patient was excluded if he/she failed to cooperate and/or had atypical fundus, or no light perception, or had sever systemic or other eye diseases.

Protocol

Based on the intention-to-screen principle, informed consent was obtained from all study participants prior to screening. The well-trained community doctor captured non-mydriatic color fundus images with a 45 field of view centered on the fovea of both eyes for each eligible participant using a Canon CR-2 Digital retinal camera. Quality of the photographs was evaluated, and the possible reasons for insufficient quality were also recorded. Fundus photographs with insufficient quality would be re-captured for another several times, and the best one would be chosen for reading. The images were uploaded to the AI grading system. The EyeWisdom (Visionary Intelligence Ltd, Beijing, China), an AI-powered fundus imaging analysis software which was integrated into an online picture archiving and communication system (PACS), enables retinal photographs upload, automated grading and generation of a single-page grading report. One pre-trained volunteer assisted the doctor to collect and input basic information into the PACS. Information including data on name, age, gender, duration of DM, medication history, etc., was matched to each picture. Visual acuity test and regular eye examination under slit lamp were also recorded. Diabetic retinopathy diagnosis was returned in three days once determined by a professional panel.

AI automated grading

The AI system named EyeWisdom was developed using a database of 25,297 fundus images (21,512 from Kaggle database, 3785 from Henan Eye Hospital and Peking Union Medical College Hospital). More than 1 million retinopathy lesions were marked manually by professional retina specialist. Based on YOLO detection system, the real-time object detection deep learning algorithm, the EyeWisdom, achieved a 90.4 sensitivity and 95.2% specificity in detecting RDR according to a pre-validation report [16]. EyeWisdom graded diabetic retinopathy severity (none, mild, moderate, severe or proliferative) according to the International Clinical Diabetic Retinopathy scale (ICDRs). Images of excellent, good and adequate quality were considered readable. Referable diabetic retinopathy (RDR) was defined as more than mild NPDR. Macular edema was not reported because it hardly could be determined simply by fundus image. The system was configured at the prefixed set point before standard DR screening. The primary outcome was the sensitivity, specificity and area under curve (AUC) for AI detecting RDR and any DR against manual standard. Severity of retinopathy in patient-wise analysis was based on the worse eye.

The reference standard

Two licensed ophthalmologists (KX and XL) with more than 10 years clinical experience were masked to the AI grading results and independently graded all the fundus color images captured from diabetes participants. An ICDR retinopathy level (scale of 0 to 4) was assigned for each image. Inconsistent reports were re-graded by a professional panel which included five senior ophthalmologists and were resolved until consensus was reached based on the majority decision.

Statistics analysis

Descriptive statistics of the characteristics of the patients were calculated. Continuous variables and categorical variables were presented as mean (SD) and n (%), respectively. R×C table was generated to calculate the κ value between AI model and panel grading. 2 × 2 table was generated to calculate the sensitivity and specificity of the AI model in RDR or ADR screening with respect to the reference standard. The 95% confidence intervals were calculated to be "exact" Clopper–Pearson intervals. Analyses were conducted using SAS statistical software, version 9.2 (SAS Institute, Inc).

Results

Total of 199 DM participants were recruited to the clinic including six who withdrew the screening. Of the 193 eligible patients, 173 (89.6%) patients had readable fundus image of at least one eye, and 20 (10.4%) patients did not have readable photographs in both eyes. The mean (SD) age was 69.3 (9.0) years with a range of 43–86 years, and among them 94 (54.3%) were women. The mean duration of DM was 11.2 (8.1) years. 19.7% (34/173) analyzed patients self-reported an unstable hyperglycemia. Figure 1 shows the STARD diagram of the participants flow according to the RDR screening by eyes.

Of the 386 included eyes, 65 (16.8%) were reported "unreadable" either by AI or manual grading. Unreadable eyes graded by AI and doctor are shown in Table 1. No significant difference ($P = 0.200$) was found about unreadable rate between AI (10.6%) and manual grading (13.0%). Total of 90 eyes (23.3%) were classified as insufficient quality images by doctor, in which 38 (19.7%) were for small pupil, 28 (14.5%) for cataract, 23 (11.9%) for both cataract and small pupil and 1 (0.5%) for cornea opacity.

The prevalence of any DR eyes based on AI screening and the manual standard was 11.0% (38/345) and 10.7% (36/336), respectively. Of patients

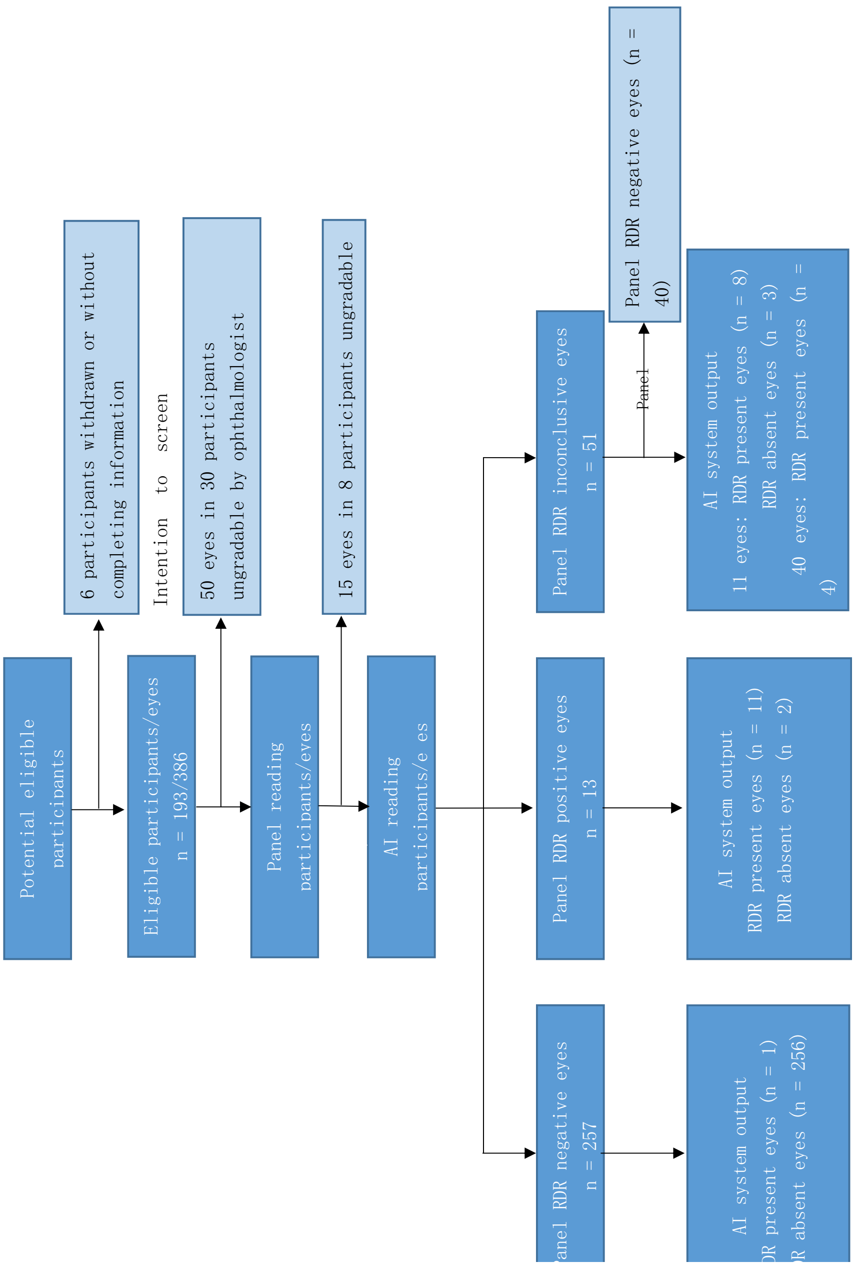

Fig.1 STARD diagram showing the flow of participants through the study

Table 1 Unreadable eye images by AI or ophthalmologist panel

AI automated grading	Reference standard		
	No DR	Unreadable	Total
No DR	0	22	22
Moderate NPDR	0	2	2
Unreadable	15	26	41
Total	15	50	65

diagnosed as any DR by AI, 54.2% (13/24) reported never receiving fundus examination previously, which was similar to the rate (55.0%, 11/20) in any DR patients diagnosed by the ophthalmologist panel. Table 2 showed detailed results of the 321 eyes (83.2%) graded both by AI and ophthalmologists. The κ value between AI model and the panel in eye image grading was 0.715.

In eye-wise analysis, the sensitivity and specificity for detection of RDR were 79.2% (95% CI: 57.9 – 92.9) and 98.3% (95% CI: 96.1–99.5), respectively. For detecting any DR, AI gave an 83.3% (95% CI: 67.2–93.6) sensitivity and 97.9% (95% CI: 95.5–99.2) specificity. The AUC for detecting RDR and any DR was 0.887 (95% CI: 0.790 –0.984) and 0.906 (95% CI: 0.834–0.979). In patient-wise analysis, the sensitivity and specificity for detection of RDR were 84.6% (95% CI: 54.6 –98.1) and 98.0% (95% CI: 94.3 –99.6), respectively. With respect to any DR detection, a 90.0% (95% CI: 68.3–98.8) sensitivity and 96.6% (95% CI: 92.1 –98.9) specificity were achieved. The AUC for detecting RDR and any DR was 0.913 (95% CI: 0.797– 1.000) and 0.933 (95% CI: 0.933 –1.000), as shown in Table 3.

Discussion

The current study recruited participants in community based on intention-to-screen principle. The results indicated that AI-assisted DR screening program was acceptable and accurate when deployed in real-world community health care clinic. Nearly 90% images were readable, and about 80% or higher RDR/any DR images were rightly detected by the program. A relatively good consistency was found between the AI system and ophthalmologists. The accuracy went better when analyzed patient-wise.

Parameters and survey items, such as image readable rate, DR prevalence, reasons of low image

Table 2 The distribution of eye grading results by AI and ophthalmologist panel

AI automated grading	Reference standard					
	No DR	Mild NPDR	Moderate NPDR	Severe NPDR	PDR	Total
No DR	279	3	2	1	0	285
Mild NPDR	3	7	2	0	0	12
Moderate NPDR	3	1	8	2	0	14
Severe NPDR	0	1	0	7	0	8
PDR	0	0	0	1	1	2
Total	285	12	12	11	1	321

Table 3 Diagnostic performance of AI for detection of RDR and any DR by eye-wise and patient-wise analysis

Grade		Sensitivity, %	95% CI	Specificity, %	95% CI	AUC	95% CI
By eye	RDR	79.2	57.9–92.9	98.3	96.1–99.5	0.887	0.790–0.984
	Any DR	83.3	67.2–93.6	97.9	95.5–99.2	0.906	0.834–0.979
By patient	RDR	84.6	54.6–98.1	98.0	94.3–99.6	0.913	0.797–1.000
	Any DR	90.0	68.3–98.8	96.6	92.1–98.9	0.933	0.933-1.000

Any DR diabetic retinopathy (any stage), RDR referable diabetic retinopathy quality and DM duration, found in this study also provided detail references in further core validation study. Van der Heijden et al. [17] firstly evaluated the IDx-DR device and found 34.5% of photographs were underutilized in practice. In our study, although 23.3% photographs were determined with insufficient quality by retina specialists, nearly 90% photographs were readable and utilized by AI. Noncompliant patients without sufficient quality photographs were mainly those aged people who have cataract. The turbid refractive stroma decreased fundus images sharpness to unreadable level for AI detection. Further screening needs to exclude those patients in the inclusion criteria. With more than 10 years of mean duration, the DR prevalence of DM patients in this community still was relatively lower than that reported in another validation of AI in Chinese city community [18] (10.7% vs. 16.1%). High education background plus standard management by primary care clinic might reduce the risks of progression into DR. A study in rural area [19] reported a 43.1% prevalence of DR in DM patients, which suggested that evaluation in rural community is mandatory.

Real-world settings validation is an important step for implication of AI programs. The FDA's Software Precertification program has pilotly regulated the validation of AI software before being certificated [20]. Van der Heijden et al. [17] firstly incorporated the IDx-DR 2.0 device in real-world clinical work flow among type-2 diabetes. Although the sensitivity of detecting RDR reached 91% according to EURO-DIAB criteria, only 68% of that was found according to ICDR criteria. The IDx-DR team launched a further prospective multicenter cohort study of 900 participants [21]. The AI system achieved a sensitivity and specificity of 87.2% and 90.7%, respectively, in screening RDR, which exceeded both the pre-specified primary endpoint goals. The study ultimately contributed to IDx-DR the first-ever approved AI system by FDA. Recently, prospective real-world validation studies are summarized in Table 4. Based on deep learning algorithm, AI program continuously improved when trained with more additional data [22]. The validation of AI-based DR screening program in community real-world setting not only acts as a post-market surveillance, but also help improve the algorithm over time as it adapts to the unique patient oriented community. Our study was a further step of certification and the first step of applying the AI to real world.

We evaluated the diagnosis performance of an AI system both by each patient and each eye. In patient-wise analysis, the final diagnosis for each participant was determined by the stage of DR of the more affected eye, which inclined to be identified by AI and manual grading. This might explain the sensitivity was better in patient-wise evaluation. Considering real-world validation adapted to the unique patient characteristics of a particular real-world site, some prospective real-world studies seemed to perform less accuracy compared with their counterparts in laboratory [17, 21]. In line with these studies, RDR diagnosis sensitivity in this study was also lower than that in laboratory (84.6% vs. 90.4%) [16]. Although the sensitivity was not as high as some published real-world results [18, 21, 23–25], the performance of our AI system overall exceeded the screening guideline recommendation (typical sensitivity and specificity > 80%) [26] and therefore was feasible in community DR screening. The main reason of lower sensitivity in this study might be the older age (mean of 69.3 years old) of the participants. Age-related eye characteristics, such as small pupil and cataract, coupled with poor compliance in image capturing would influence the picture quality and accordingly performance of AI reading.

Notably, the ensemble AI system was shown to have excellent sensitivity, specificity and AUC in detecting any DR both by eye-wise and patient-wise analysis, especially the high specificity (> 95%). These results were very close to studies validated under experimental [14] and real-world scenario [18, 23]. It evidenced that the AI system was promising in early stage of DR detection and non-DR could also be rightly distinguished among DM patients. However, we believe that current AI product should play a complementary role. Anyway, for those patients with normal but atypical fundus, false-positive results would increase their medical cost once being falsely referred to eye hospital. Fundus image of DM patient identified as any DR or RDR by AI algorithms should still be reviewed by an ophthalmologist before a referral is made to avoid false positive.

Conducting screening in workforce population to eliminate age-related factors might be a method of improving sensitivity. In our study, although the mean age was old, pictures were not particularly selected in

Table 4 Summary of prospective real-world validation studies of AI systems

Study	Country	AI software	Validation data set	Grading scale	RDR detection		Any DR		AI-readable DR prevalence rate	
					sensitivity	Specificity	Sensitivity	Specificity		
Our study	China	YOLO	1 primary diabetes care hospital, 166 patients	ICDRS	84.6%	98.0%	90.0%	96.6%	89.40%	ADR eye: 10.7% (36/336)
He et al. 2019	China	Inception-V4	1 Community Hospital, 889 patients	ICDRS	91.2%	98.9%	90.1%	98.5%	–	ADR patients: 16.1% (143/889)
Natarajan et al. 2019	India	Inception-V3	community-based, 231 patients	ICDRS	100.0%	88.4%	85.2%	92.0%	92.20%	ADR patients: 12.7% (27/213)
Rajalakshmi et al. 2018	India	EyeArt™	1 tertiary care diabetes hospital, 296 patients (dilated)	ICDRS	99.3%	68.8%	95.8%	80.2%	98.30%	RDR patients: 48.0% (142/296)
Abramoff et al. 2018	USA	IDx-DR	10 primary care sites, 892 patients	ETDRS	87.2%	90.7%	–	–	96.10%	RDR patients: 23.8% (198/819)
Vanderheijden 2017	USA	IDx-DR	the Hoorn DCS center	ICDRS	68.00%	86.0%	–	–	66.3%%	RDR patients: 8.1% (73/898)
				EURODIAB	91.00%	84.0%	–	–		
Kanagasingam et al. 2018	Australia	Inception-v3	1 primary care practice. 193 patients	–	–	–	2 DR were identified	92.00%	–	RDR patients: 1.0% (2/193)

order to evaluate a true scenario of AI in screening DR. Besides, as a dilated pupil is helpful in improving the quality of retina image and makes DR characteristic more visible, it accordingly contributes to a higher sensitivity. Rajalakshmi et al. [24] validated a smartphone-based AI system among 296 DM patients with dilated pupils at a tertiary care hospital. A 99.3% sensitivity were exhibited in detecting RDR, while the specificity was only 68.8% which explained that more non-DR patients were misdiagnosed as DR. In our study, likewise most of current studies, pupil dilation was not applied for safety consideration of old age people and for screening purpose. Further, training the AI system with more real-world data containing various characteristics is also necessary to improve its performance.

The key strength of our study was its prospective real-world design located in primary care hospital. Stronger evidence could be expected about the practical performance of our AI system. Participants were pre-recruited by community medical workers, and therefore, it was more efficient in fast targeting DM patients. The primary limitation of this study might be its small sample size. Coupled with relatively low DR prevalence, we only evaluated the AI performance in detecting RDR and any DR. Besides, this pilot study was limited to only one community with relatively higher educated population which might undermine its generalizability. Still, macular edema was not reported in our study because fundus images is not sensitive in detecting this condition; this might miss some RDR patients according to ICDRS criteria.

In summary, this community-based evaluation of AI-assisted DR screening system among aged DM patients showed high image readable rate, high specificity and acceptable sensitivity for identifying RDR and any DR by eye-wise or patient-wise analysis. The results provide information for further multi-site studies with more participants aiming at a comprehensive evaluation of each DR grade, even an analysis stratified by different characteristics such as the duration of DM.

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Author contributions SM helped in protocol development, data management, analysis, results interpretation and manuscript writing. KX contributed to data collection, fundus image reading and results interpretation. XL, ZZ, SL and XJ were involve in fundus image reading, ZZ, YY, XJ and BL contributed to program coordination. BL helped in language proofreading, manuscript revision and funding support.

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Data availability The data used to support the findings of this study are available from the corresponding author upon request.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval The study was approved by the Ethical Committee of Henan Eye Hospital.

Informed consent Informed consents were obtained from the subjects after explanation of the nature of the study.

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